# Erik Hill

**AI Evaluation & Testing Engineer**

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## SUMMARY

Self-taught AI evaluation and testing engineer building deterministic, replayable systems for measuring coding agents and LLM applications.

**Merged upstream at promptfoo (by its co-founder), GitLab and TeaVM**; an arXiv preprint; **30+ public repos, three PyPI packages, one npm package**; a game shipped to Google Play. I build **verifiable-evaluation systems** whose claims are checked rather than asserted: **a public registry of 12 accepted evaluation claims, each hash-pinned to a commit and replayed weekly in CI** (11 of 12 replayed green on 2026-10-19): deterministic, no LLM judges, failures published by design. Every upstream fix, with its proof: erikhill.dev/findings

## SKILLS

**Evals & AI:** No-LLM-judge grading · Mutation testing · Differential-oracle testing · Property-based testing · Regression gating (nDCG) · Pre-registration & minimum-detectable-effect analysis · RAG · LLM application development · Tool-calling & guardrails · Safety (OWASP LLM Top 10, human-in-the-loop) · Prompt engineering · Context engineering · Model Context Protocol (MCP) · LangGraph · Multi-agent orchestration
**Backend & data:** Python · FastAPI · Pydantic · SQLAlchemy 2.0 · PostgreSQL · pgvector · REST / OpenAPI
**Frontend / mobile:** TypeScript · JavaScript · React · Next.js 16 · Java · Capacitor · Android (adb) · WebAssembly (Pyodide) · TeaVM (JVM to JavaScript)
**Ship & verify:** pytest · Vitest · JUnit · jqwik · Playwright · PyPI / npm packaging · GitHub Actions (matrix CI) · Cloudflare Pages (Wrangler) · Neon · Test automation · QA for LLM systems

## SELECTED PROJECTS, all public at github.com/egnaro9

• **agent-certlab**: **Two real coding agents certified, every verdict regradeable from artifacts**. Three **seeded-defect task families** with provable answer keys (incl. **coordinated two-file defects** no single-file patch can pass), a **precondition prover** (clean passes, every seed fails; it refused one of my own defects), and **calibration agents** the harness must separate before certifying.

• **reference-fleet**: **An answer key for eval suites**: six models each broken **one documented way at an exactly reproducible seeded rate**; a naive suite detects **1 of 6** (paired protocol). A **LoRA-trained native defect** showed training's rate control is lossy: mixture 0.507 → realized **0.20 greedy, 0.31 at temp 1.0**.

• **vac-protocol**: **Verifiable Agent Claims**: evidence-bundle format + **offline verifier** behind the registry above, with a **public record of it rejecting my own evidence**. Plus **vac-gate**: no verified contract, no green check.

• **evalmut**: **Mutation testing for eval graders** (**PyPI**, with its **gradecore** engine): inject known defects, report which checks stayed green. **No LLM judge**; every operator tied to a documented real failure. Found **4 holes in gradecore's own graders** and **8 in ports of a widely-used framework's assertions**; fixed **three fail-open graders** in gradecore 0.10.3. *325 tests.*

• **deadgate**: **a linter for CI checks that cannot fail** in GitHub Actions workflows (**PyPI**). Across **275 repositories** I hand-labelled all 653 of its findings and **deleted two of its four detectors** that the labels showed had never once been right: **36.1% false, measured rather than estimated**. Found an unguarded required status check in **heygen-com/hyperframes**, fixed by my merged PR #5182. *228 tests.*

• **deadcheck**: **does an eval rubric catch what it claims?** Run on **three rubrics I did not write**: OpenAI evals' `Match` accepts "10" for ideal "1"; a promptfoo example passes refusal-then-compliance. Seven probe rounds found **23 defects in deadcheck itself under a passing suite**. *185 tests.*

• **rag-eval-lab**: deterministic RAG **eval harness** that catches a planted hallucination in CI; from-scratch **BM25 matching the published SciFact baseline (0.664 vs 0.665)**. *60 tests.*

• **agent-graph**: a **LangGraph ReAct agent** with **unit-testable** multi-step tool use: a swappable mock/LLM policy runs it deterministically without a key; a **safe AST calculator** (no `eval`); a max-step guard. *67 tests.*

• **harness-builder**: **More scaffolding scored worse, for 22x the cost**. Three harness shapes on one 20-task suite, **zero errors in 60 runs**: one drafter 95%, planner plus drafter 90%, four-call panel with judge 80%. The panel won **0 of 20 tasks**, and the tool reported **"this suite cannot decide"** rather than ranking.

• **tapdodge-engine**: **One rulebook, two runtimes, and a check that fails when they stop agreeing**: a JVM-versus-browser (TeaVM) differential in CI. In a committed fault injection it was the **only** check that caught a runtime-supplied-RNG fault; the same-runtime determinism test and the golden file both passed it.

• **Also public: exposure-watch** (AI agents vs. plain code at known-exploited-vulnerability triage; pre-registered, 57 held-out pairs: a **tie on correct answers**, 47 to 45, p = 0.69, but the agent pipeline sent its 10 unsure cases to a person and got **0 wrong**, vs. 12 wrong from code plus a keyword rule) · **mcp-tools** (MCP server, no SDK) · **pi-gates** / **pi-eval** (fail-closed commit and model gates, plus no-LLM-judge grading, for the pi coding agent) · **llm-wire-stub** (npm) · **readme-test-count** (fails CI when a README's test count disagrees with the suite; runs in the CI of six of my repos)

## REVIEWED BY OTHERS

• **Two pull requests merged into promptfoo** (26k stars), both merged by its co-founder. #11278: a blank BLEU reference, such as an unset template variable, was scored instead of ignored; the fix shipped in promptfoo 0.124.0. #10256: the `rouge-l` / `rouge-s` assertion types, merged 2026-10-09.

• **Three merge requests merged into GitLab**: two CVE records whose stored score contradicted their own published vector (CVE-2023-5009 stored 9.6 CRITICAL; its vector computes 8.2 HIGH), and the security handbook's disclosure window, corrected from 30 days to 90.

• **External audit of vac-protocol:** an outside engineer falsified my own verifier, cheapest forgery **four bytes**. Fix merged, then I found the same defect class **four more times** (arXiv, below). As **sole maintainer** I merged four of his pull requests and answered his four specification issues (protocol v0.2); on two I showed his proposed fix would not close the hole and shipped a wider one.

• **Peer review with Bartłomiej Nawara**, senior AI/ML engineer: his missing-not-at-random critique shaped **model-drift**'s **reliability floor** (credited in the repo); I found **two design faults** in his pre-registered study.

• **Closed a bug report open in TeaVM since 2018** that its own suite could not catch: a **one-character error** in its GregorianCalendar emulation, found by differential-testing the browser port against the JVM. The patch re-enables the four commented-out assertions covering that line; merged same day by the maintainer.

## PUBLICATIONS

“Four Ways to Forge a Bundle My Own Verifier Calls Clean: Refusal-Site Mutation Testing of an Evidence-Bundle Verifier.” Erik Hill. **arXiv preprint, 2026**. arXiv:2608.26183
**25+ articles** on evaluation and testing failures. dev.to/agentdev9 · **Drift Notes**: erikhill.dev/notes

## EXPERIENCE

**AI/ML Engineer · Epix Devo LLC** · *Contract · Oct 2026 to Present*

**Independent AI Evaluation & Testing Engineer · Solo Practice** · *Remote · Mar 2026 to Present*

- Designed **five roles across a four-stage production loop** (Strategy → Execution → Critic → Ops) that plans, builds, reviews, and validates changes to a real React / Capacitor product, human-in-the-loop on every consequential step, plus an **asynchronous Evaluation harness** that scores completed batches.
- Built per-role model routing with manager-driven handoffs: each stage runs as a fresh subagent pinned to the model its role needs; enforced rules with **property-based invariants** (jqwik) over a native engine and **adb-driven on-device validation**. Case study: exposure-watch/docs/HARNESS.md
- **Operate model-drift as a running service, not a demo**: its scheduled probe has run every calendar day since 2026-07-19 on a public board; the **frozen hash-fingerprinted suite** now covers **17 models / 5 labs**, with a derived **minimum-detectable-regression floor**.
- Diagnosed an apparent **66-point overnight drop** on that board (69% to 3% on a frozen suite) as a **provider rate cap rather than model drift**: 34 of 35 calls hit a 429 against a 30 RPM shared tier. Shipped Retry-After-aware retries, a per-host throttle and a reliability floor, so a rate limit can no longer render as a regression.

**Indie Game Developer · SeraphLight Studios** · *Remote · Nov 2025 to Present*

- **Shipped Tap Dodge Rush to Google Play** (Kotlin / Java, Android Views + Canvas, AdMob) after closed testing with 12+ testers and store review. A second, larger game (React / Capacitor), the product the harness above operates on, is in development.

**Kitchen Manager / Shift Lead · Big Island Brewhaus & other high-volume kitchens** · *Hawaii · 2018 to 2025*

## EDUCATION

**Clark College**, 2007 to 2009 (some college, no degree).